
Input: Encoding circuit C ; address qubit set A ; data qubit set D ; cut budget $k_{\max}$.

Output: Cut set $\mathcal{C}$ of two-qubit gates to replace by QPD cuts.

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1  $G_{\text{cross}} \leftarrow \emptyset$ ;  
2 foreach two-qubit gate  $g = (q_i, q_j)$  in  $C$  do  
3   if  $(q_i \in A \wedge q_j \in D) \vee (q_i \in D \wedge q_j \in A)$  then  
4      $d \leftarrow |i - j|$ ;  
5      $G_{\text{cross}} \leftarrow G_{\text{cross}} \cup \{(g, d)\}$ ;  
  
6 sort  $G_{\text{cross}}$  by  $d$  in descending order ; //  $\mathcal{O}(|G| \log |G|)$   
7 return the first  $\min(|G_{\text{cross}}|, k_{\max})$  gates of  $G_{\text{cross}}$  as  $\mathcal{C}$ 
```
